

Dark Roast Design System — v1.0.0-draft

Full specification. Extends `theme-context-cache-v4.json` (colors only) with primitives, components, and app-specific tokens.

Status legend:

-  **VERIFIED** — sourced from canonical Dark Roast v2.0.0 artifacts
-  **DEFAULT** — opinionated proposal from Claude, pending your approval
-  **TBD** — needs real-world measurement or decision

Review workflow: Scan each section. Mark DEFAULT items you approve, reject, or want revised. Once locked, values migrate into the canonical JSON and generate platform targets via `dark-roast.py`.

1. Color

Color tokens are already canonical. Summarized here for self-containment — see `theme-context-cache-v4.json` for full provenance.

Surface Scale (6 steps, darkest → lightest)

Token	Hex	Use
<code>void</code>	<code>#120C06</code>	Page background
<code>obsidian</code>	<code>#160E08</code>	Content background
<code>dark_cacao</code>	<code>#1E140E</code>	Elevated surfaces (modals, sheets)
<code>espresso</code>	<code>#2A1C14</code>	Card backgrounds
<code>roasted_bean</code>	<code>#3C2A1E</code>	Borders, dividers
<code>crater</code>	<code>#4D3B31</code>	Geological accent, card bottom stripes

Foreground Tiers

Token	Hex	Use
display	#FFF7EE	Hero text, brand chrome only
workhorse	#D4C4A8	Terminal, code, body reading
secondary	#8B7355	Muted body, labels
tertiary	#465945	Timestamps, metadata
warm_white	#F0E6D0	Bright callouts, ANSI 15
bone	#EBE1D7	Codex ink, midpoint

Action / State

Token	Hex	Use
accent	#E69A4C	Primary accent, cursor, CTAs
accent_hot	#D2691E	Gradient terminal, worsening state
accent_muted 	#C07A4A	Reader contexts (Mystic2 parity)
success	#4CC4B4	Success, live data, kinetic teal
warning	#BFA162	Warning, caution
error	#C75B39	Error, terminal error
scarlet	#C44C4C	Clinical critical severity
gold	#DAA520	Clinical stable severity

 **Proposed addition:** `accent_muted #C07A4A` — imported from Mystic2 tuned. Gives the system a quieter amber for reader contexts where full `accent` is hostile. Reject if you want to keep the two systems visually distinct.

Crater Brown Layer

`crater_primary #3C2A21` , `crater_light #4D3B31` , `asparagus #465945` , `rustic #480404` ,
`rose #480607`

2. Opacity

Alpha scale for overlays, glows, scrims. Hex-suffix form.

Token	Decimal	Hex	Use
0	0.00	00	Transparent
4	0.04	0A	Subtle divider
8	0.08	14	Hover overlay
12	0.12	1F	Amber glow
15	0.15	26	Teal glow
24	0.24	3D	Disabled state
40	0.40	66	Modal scrim
60	0.60	99	Heavy scrim
85	0.85	D9	Export modal backdrop
100	1.00	FF	Opaque

Rationale: Scale ties to existing glow tokens (`glowAmber 0.12` , `glowTeal 0.15` from Swift file). Fills gaps for hover (8%), disabled (24%), modal backdrop (85% — matches Mystic2 export panel).

3. Spacing

4px base unit. Standard iOS/SwiftUI 4pt grid.

Token	px	rem	Use
0	0	0	No spacing
0.5	2	0.125	Hairline gap
1	4	0.25	Tight intra-component
1.5	6	0.375	Chip padding

2	8	0.5	Related element gap
3	12	0.75	Standard content padding
4	16	1.0	Card padding, section spacing
5	20	1.25	Larger card
6	24	1.5	Section gap
8	32	2.0	Major section
10	40	2.5	Page-level
12	48	3.0	Hero
16	64	4.0	iPad page margins
20	80	5.0	Large hero
24	96	6.0	Extreme hero

4. Radii

Token	px	Use
none	0	Full-bleed
xs	2	Inline chips, tight badges
sm	4	Tight controls
md	8	Standard buttons, inputs
lg	12	Cards (matches <code>DarkRoastCard</code> Swift modifier)
xl	16	Large cards, modals
2xl	24	Hero cards
full	9999	Pills, circular avatars

Anchor: `radii.lg = 12px` is locked to your existing `DarkRoastCard` modifier. Everything

else scales around it.

5. Elevation

Dark Roast elevation uses atmospheric bleed + subtle shadows + border-weight variation.
No iOS-typical hard drop shadows.

Token	Spec	Use
<code>flat</code>	no shadow, 1px hairline border	Inline elements
<code>raised</code>	<code>0 2px 8px rgba(0,0,0,0.4) + hairline</code>	Standard cards
<code>floating</code>	<code>0 8px 24px rgba(0,0,0,0.5) + amber glow</code>	Modals, popovers, hover
<code>critical</code>	<code>0 8px 24px rgba(0,0,0,0.6) + scarlet glow</code>	Critical alerts, high-severity patient
<code>live</code>	<code>0 4px 16px rgba(0,0,0,0.4) + teal glow</code>	SSE-streaming cards

Key insight: Glow color = semantic state. `floating` means “user interacting.” `critical` means “something’s wrong.” `live` means “data moving.” The shadow alone shouldn’t carry meaning — pair with glow.

6. Typography

Families are **VERIFIED**. Size/weight/line-height ramps are **DEFAULT**.

Families

Role	Family	Weights	Use
<code>display</code>	Playfair Display	700, 900	Hero, brand, app titles
<code>heading</code>	Instrument Sans	400–700	Patient names, track titles, card headers

body	DM Sans	400, 500, 700	Clinical descriptions, UI labels
mono	Fira Code	400–600	Lab values, ISRC, JSON, vitals, code. Ligatures enabled.

Scale 🎯 (1.25 minor-third ratio)

Token	px	line-h	Weight	Use
xs	10	1.4	500	Badges, micro-labels
sm	12	1.5	400	Metadata, timestamps
base	14	1.6	400	Body copy, UI default
md	16	1.5	400	Readable body
lg	18	1.4	500	Card titles
xl	22	1.3	600	Section headers
2xl	28	1.2	700	Page titles
3xl	36	1.15	700	Hero subheads
4xl	48	1.1	900	Hero display (Playfair)
5xl	64	1.0	900	Marketing hero only

Semantic Roles 🎯

Instead of naming type by size, name by job:

Role	Family	Scale	Color
hero	display	4xl	foreground.display
page_title	heading	2xl	foreground.display
section	heading	xl	foreground.workhorse
card_title	heading	lg	foreground.workhorse
body	body	base	foreground.workhorse

body_muted	body	base	foreground.secondary
label	body	sm	foreground.secondary
metadata	body	sm	foreground.tertiary
code	mono	base	foreground.workhorse
lab_value	mono	md (500)	foreground.workhorse
timestamp	mono	xs	foreground.tertiary
vital_sign	mono	lg (600)	foreground.workhorse

Rationale: `lab_value`, `vital_sign`, `timestamp` are specific to somaCURA clinical contexts. Lab values use mono at 16px/500 — legible at a glance during rounds.

7. Motion

Tuned for calm/deliberate. Avoids bouncy springs except for delight moments.

Durations

Token	ms	Use
instant	0	<code>prefers-reduced-motion</code>
fast	120	Hover, tap feedback
base	200	Standard transitions
moderate	320	Card expand, tab switch
slow	480	Modal enter
deliberate	640	Hero reveal
ambient	1200	Breathing glows

Easings

--	--	--

Token	Curve	Use
<code>ease_out</code>	<code>cubic-bezier(0.16, 1, 0.3, 1)</code>	Most enters
<code>ease_in</code>	<code>cubic-bezier(0.7, 0, 0.84, 0)</code>	Most exits
<code>ease_in_out</code>	<code>cubic-bezier(0.65, 0, 0.35, 1)</code>	Symmetric
<code>spring_snappy</code>	<code>.spring(response: 0.35, dampingFraction: 0.7)</code>	Confirmations
<code>spring_soft</code>	<code>.spring(response: 0.55, dampingFraction: 0.85)</code>	Card expansion

Named Keyframes

- `breathe_amber` — live indicator, streaming cursor (1200ms `ease_in_out`)
- `pulse_scarlet` — critical patient, flagged lab (1200ms `ease_in_out`)
- `shimmer_skeleton` — loading, pre-stream (1500ms linear)

8. Z-Index 🎯

Never hand-write arbitrary z values.

base 0 → raised 1 → sticky 100 → header 200 → dropdown 300 →
overlay 400 → modal 500 → popover 600 → toast 700 → tooltip 800 → debug 9999

9. Icon Sizing 🎯

Token	px	Use
<code>xs</code>	12	Inline in text
<code>sm</code>	14	Badge icons
<code>md</code>	16	Standard buttons
<code>lg</code>	20	Card-level, toolbar

x1	24	Primary CTA
2x1	32	Hero icons
3x1	48	Feature illustrations

Stroke widths: thin 1.5, normal 2, bold 2.5.

10. Components

Built from primitives above. All token references use `category.token` form.

Button

- **primary** — `accent` bg, `void` fg, `md` radius. Hover → `accent_hot`.
- **secondary** — transparent bg, `accent` fg, 40% `accent` border.
- **ghost** — transparent bg, `workhorse` fg, `espresso` hover.
- **danger** — `error` bg, `warm_white` fg, hover → `scarlet`.
- Sizes: `sm` / `md` / `lg` via padding + `font_scale`.

Card

- **base** — `espresso` bg, hairline border, `lg` radius, `raised` elevation, crater bottom stripe.
- **elevated** — adds `floating` elevation + amber glow.
- **critical** — `scarlet` border + critical elevation + `scarlet` glow.
- **live** — teal border + live elevation + teal glow.

Input

- Base: `obsidian` bg, `roasted_bean` border, `accent` focus border + 24% `accent` ring.
- Invalid: `error` border + 24% `error` ring.

Nav

- `obsidian` bg, `roasted_bean` bottom border, `secondary` inactive / `accent` active, 2px bottom indicator.

Modal

- 85% black backdrop, `dark_cacao` bg, `roasted_bean` border, `x1` radius, floating elevation. Enter `slow/ease_out`, exit `moderate/ease_in`.

Toast, Chip, Divider, Tooltip

See JSON for full specs. All derive from primitives — no hand-tuned values.

11. State Variants

Applied to any interactive token.

- **hover** — 8% white overlay, `fast` transition
 - **active** — 12% black overlay, 0.98 scale, instant
 - **focus** — 24% accent ring, no outline
 - **disabled** — 40% opacity, not-allowed cursor
 - **selected** — 15% accent bg, full accent border
-

12. Accessibility

Reduced Motion

`prefers-reduced-motion: reduce` → all durations become `instant`, ambient keyframes freeze at 70% opacity.

High Contrast

`prefers-contrast: more` → upgrade all foreground tiers one level (`workhorse` → `warm_white`, `secondary` → `workhorse`, etc.) and strengthen divider visibility.

Contrast Targets

Minimum WCAG ratios per role, actual against `void`:

Role	Min	Actual
display	12.0	17.4 

workhorse	7.0 (AAA)	11.3 
secondary	4.5 (AA)	5.1 
tertiary	3.0	3.2 
accent	4.5 (AA)	8.4 
metadata_ignore	1.8–3.0	— (Mystic2 IGNORE pattern)

Imported from Mystic2: the `metadata_ignore` role — legible-but-ignorable layer with explicit contrast *ceiling*, not just floor. Design primitive for “eye should skip this.”

13. Print / Export

Dark Roast doesn’t degrade to light substrates — explicit inversions needed.

Dark token	Paper substitution
void	#FFFFFF
obsidian	#FAFAF8
espresso	#F5F2ED
workhorse	#2A1C14
display	#120C06
accent	#B26540
error	#A23818
success	#2A8778

Rules: amber drops 20% saturation (ink bloom), scarlet → burgundy on paper (severity preserved).

14. somaCURA

Severity Mapping 

State	Color	Use
critical	scarlet #C44C4C	Deteriorating, ICU-eligible, panic labs
worsening	accent_hot #D2691E	Trending worse
improving	accent #E69A4C	Trending better
stable	gold #DAA520	Stable monitoring
resolved	success #4CC4B4	Archived

Pipeline States 🎯

Visual treatment per pipeline stage (Clinical Update → HyperDrive NER → Problem Approval → SSE Stream → Complete). Each has color + SF Symbol icon + optional animation. See JSON.

Components 🎯

- **census_row** — base `espresso`, severity overlay 8% on critical/worsening, 3px left accent border when selected, 8px severity dot before name, mono vital signs
- **problem_card** — 4px severity stripe on top, sparkline in severity color, 32px sparkline height
- **flagged_lab** — 8% scarlet bg, 40% scarlet border, pulse animation
- **sse_stream_cursor** — 2px amber bar with breathe animation
- **problem_list_item** — 6px severity dot prefix, resolved items demote to tertiary fg (**no strikethrough** — per your clinical doc preferences)
- **disposition_barrier** — dashed 40% warning border, warning triangle icon

15. Song Expanse ✅🎯

Token Separation (CRITICAL RULE)

This is the architectural contract. Violating it breaks the whole model.

Chrome = AppTheme (Dark Roast tokens from this spec). Nav, tabs, modals, settings, controls. **Constant across songs.**

Content = TextColors (9-tier palette derived from album art). Lyrics, annotations, timeline, song story, credits. **Album-responsive**.

Never mix. The harmonizer (`dark-roast.py harmonize`) scores TextColors against the theme collection for chrome-content coherence.

Vinyl Renderer

Complete mapping from canonical Dark Roast v2.0.0:

- `case_bg` = void, `platter_shadow` = `crater_primary`, `tonearm_metallic` = secondary fg, `tonearm_tip_glow` = accent, `play_button` = accent, `loading` = success (teal), `marquee_scrim` = `crater_primary`, `marquee_grid` = 12% accent, `timeline_rail` = `crater_primary`, `timeline_thumb` = accent

TextColors Generation Rules

9-tier palette derived from album art:

- `textColor1–3`: content foreground tiers (target ≥ 7.0 / ≥ 4.5 / ≥ 3.0 contrast vs chrome bg)
- `textColor4–6`: accent tiers (lyric highlight, annotation, credits)
- `textColor7–9`: content surfaces (base, elevated, dim)

Generation flow:

1. Extract dominant + accent colors from album art (k-means or perceptual clustering)
2. Reject palettes where fewer than 3 tiers meet contrast targets against Dark Roast void
3. If fail → fall back to Dark Roast foreground tiers
4. Cache per songId, invalidate on art change
5. Harmonizer pass: if clash against current chrome, dim accent tiers 15%

Content Components

- **lyric_line** — active/played/upcoming opacity cascade, `textColor4` for annotations
- **song_story_chunk** — `textColor7` bg, `textColor1` body, `textColor5` enrichment accent
- **credits_bar** — `textColor9` bg, metadata typography
- **artist_intel** — rotating facts, deliberate $\times 10$ interval
- **career_timeline_heatmap** — 5-stop intensity scale from `textColor9` → `textColor4`, `textColor5` for annotated moments

16. Additional Apps 🕒

- **Dictum** — cross-platform voice dictation (TBD component specs)
- **tradingDesk** — financial analysis webapp (TBD component specs)
- **Narwhal2 / Mystic2** — see `theme-context-cache-v4.json`
`related_themes.mystic2_tuned` . Uses Narwhal2 app semantic model, not this component spec.

17. Generation Pipeline

This spec drives everything downstream. Never hand-edit generated files.

Target	File	Generator
Web CSS	<code>tokens.css</code>	<code>dark-roast.py export css</code>
Swift	<code>DarkRoastTokens.swift</code>	<code>dark-roast.py export swift</code>
iTerm	<code>dark-roast.itermcolors</code>	<code>dark-roast.py export item</code>
Alacritty	<code>dark-roast.toml</code>	<code>dark-roast.py export alacritty</code>
Warp	<code>dark-roast.yaml</code>	<code>dark-roast.py export warp</code>
Codex Desktop	<code>import string</code>	<code>source from spec.color + typography + action</code>

Validation:

- `dark-roast.py validate` — WCAG + OLED wake
- `dark-roast.py harmonize` — TextColors coherence scoring

Review Checklist

Work through these in order. Each 🎯 item is a yes/no/revise decision.

- §1 Color — `accent_muted #C07A4A` addition (Mystic2 import)
- §2 Opacity — scale granularity sufficient?
- §3 Spacing — 4px base + scale granularity
- §4 Radii — 8 steps including `radii.lg = 12px` card anchor
- §5 Elevation — 5 states (flat/raised/floating/critical/live) + glow tokens carry semantic meaning
- §6 Typography — 1.25 ratio scale + 12 semantic roles
- §7 Motion — 7 durations + 5 easings + 3 named keyframes
- §8 Z-Index — 11-step named scale
- §9 Icons — 7 sizes
- §10 Components — button/card/input/nav/modal/toast/chip/divider/tooltip all derive from primitives
- §11 State Variants — hover/active/focus/disabled/selected
- §12 Accessibility — reduced-motion, high-contrast, contrast-target bands including `metadata_ignore` (Mystic2 import)
- §13 Print/Export — inversions with saturation drop for amber, scarlet→burgundy
- §14 somaCURA — pipeline states + `census_row` + `problem_card` + `flagged_lab` + `sse_stream_cursor`
- §15 Song Expanse — TextColors generation rules + 5 content components
- §16 Additional apps — prioritize Dictum or tradingDesk next?

Reject, revise, or approve each. Once locked, the JSON becomes canonical and regenerates all platform targets.